Nasa Ozone and Temperature Profiles Dashboard

The NASA sub-orbital order tool (SOOT) gives access to the airborne and field campaign data archived at the Atmospheric Science Data Center (ASDC).

https://asdc.larc.nasa.gov/soot/browse

The ozone layer is a region of Earth's stratosphere that contains a high concentration of ozone (O_3) relative to the rest of the atmosphere. It acts as Earth's natural sunscreen by absorbing most of the Sun's medium-frequency and high-frequency ultraviolet radiation. However, ground-level ozone in the troposphere can be dangerous because it acts as a harmful air pollutant created by human activity. The tropopause is the atmospheric boundary layer that separates the troposphere from the stratosphere. Therefore, this dashboard was made to improve on accessibility and readability of NASA campaigns data related to Earth's ozone and temperature. It aims to facilitate research in a classroom setting by allowing students to view interactive visualizations made from interesting datasets. This dashboard was built by expanding on the work done by a previous team using their website as a starting point.

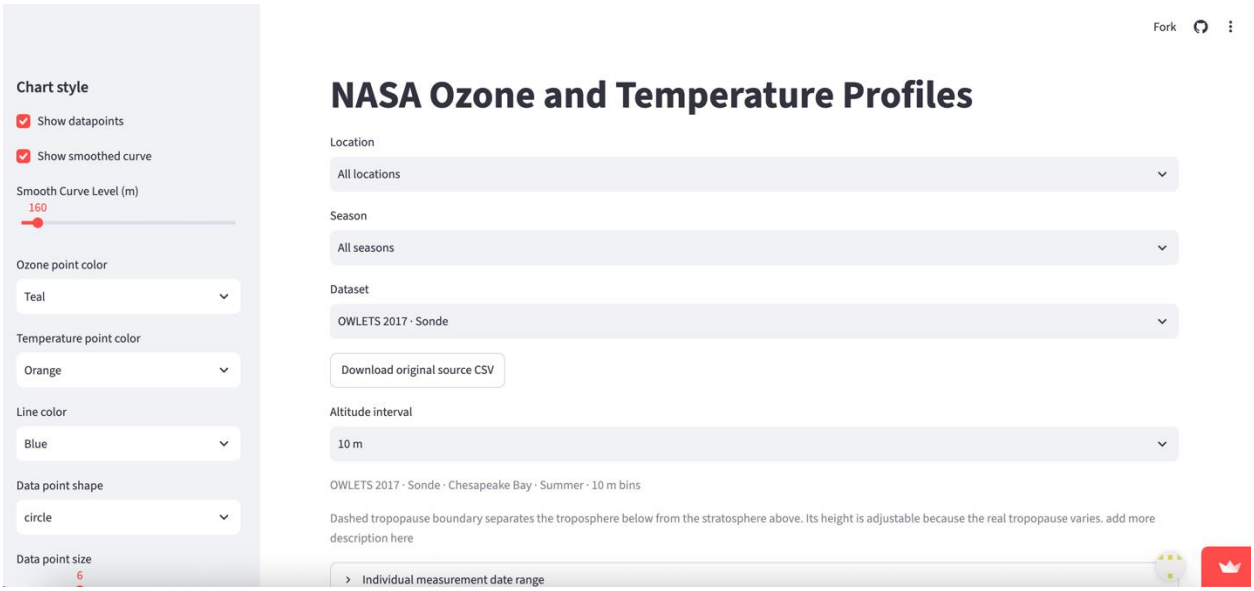
Nasa Ozone and Temperature Profiles Dashboard created by Virginia Reider, Raphael Doh-Nani, and Niurika Gonzalez

Previous Student Website:

NASA SOOT—ICARTT Downloader + CSV Converter
A tool developed in Spring 2026 by Grant Bailey, Maggie Gonzalez, Greyson Willhite to convert NASA SOOT Data to csv format.

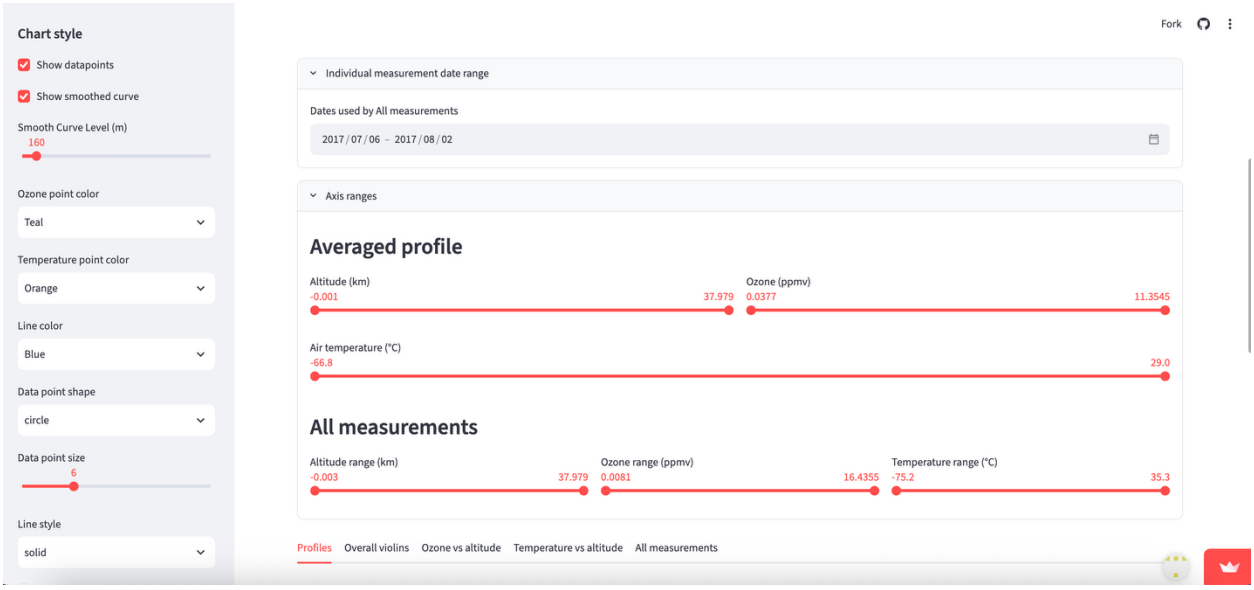
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When first entering the dashboard website, there are drop down menus allowing the user to select the location, season, and available dataset. Under these three selections, there is a button that gives the user the ability to download the original dataset source CSV.

Next the user can choose whether to view the data with an altitude interval of 10 m, 50 m, 100 m, or 500 m.



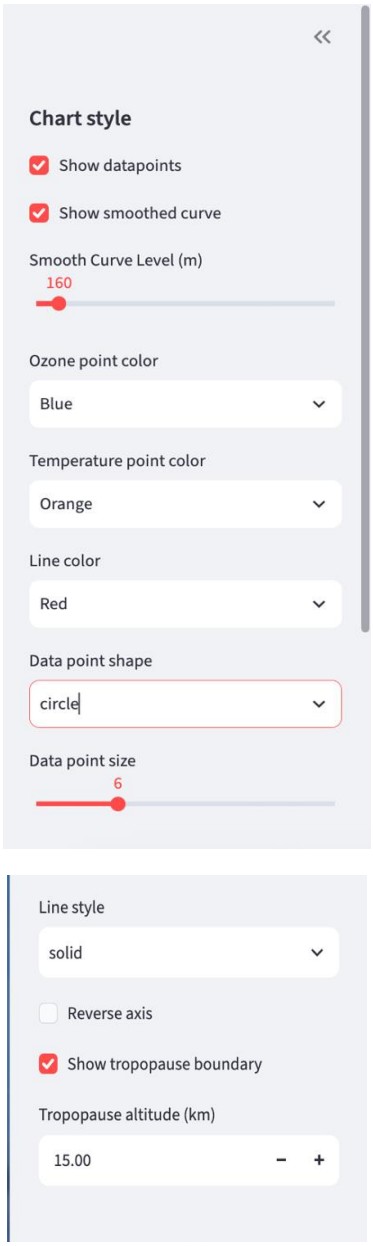
A specific date range can be chosen depending on when the data was recorded. This will change the timeframe of the data points in the corresponding plots below.

The user also can change the axis ranges of the average profile and all measurements.

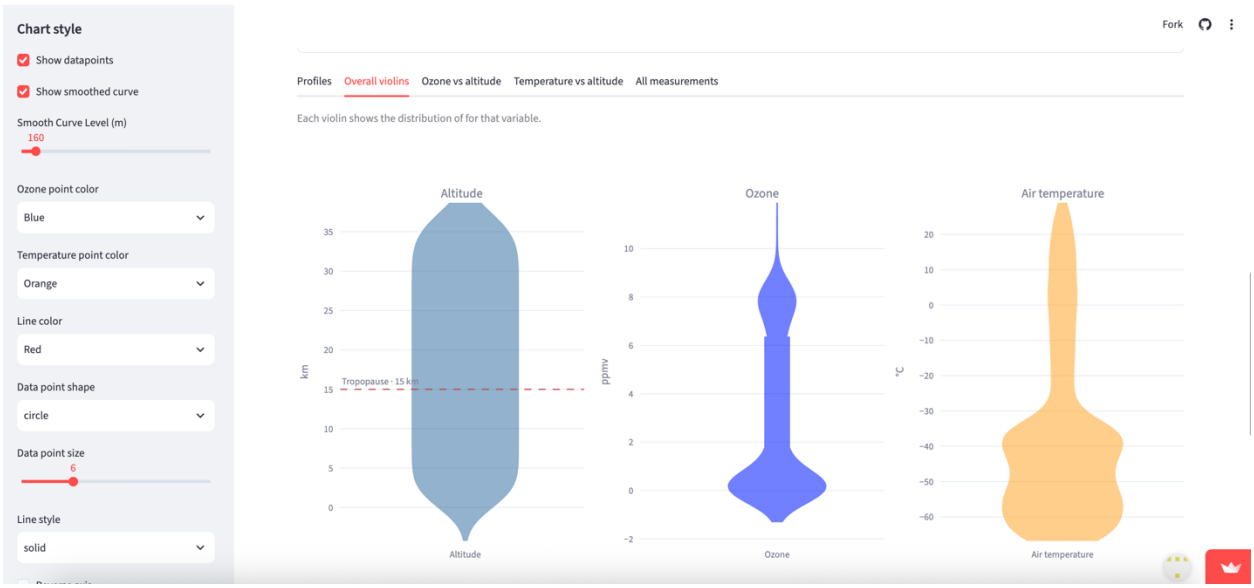


Below the menu selections the user can view various charts of the selected data including the profiles, overall violins, ozone vs altitude, temperature vs altitude (if that information is available for the specific dataset), and all measurements together.

The profiles chart shows ozone profile means data points along with an ozone profile line. It shows ozone in ppmv along the x-axis and altitude in km along the y-axis. Some datasets also show air temperature profile means data points along with an air temperature profile line. It shows air temperature in Celsius along the x-axis and altitude in km along the y-axis. These charts also show a tropopause line that can be manipulated or removed by the user with the menu bar on the left.



The menu bar on the left side allows the user to change the style of the profiles chart. They can choose to remove or show datapoints and the smoothed curve. They can also change the smooth curve level. The user can personalize the chart to their liking by changing the point color, line color, data point shape, data point size, and line style. They can also choose to reverse the x and y axis. There is an option to show the tropopause boundary line and to change its altitude measurement.

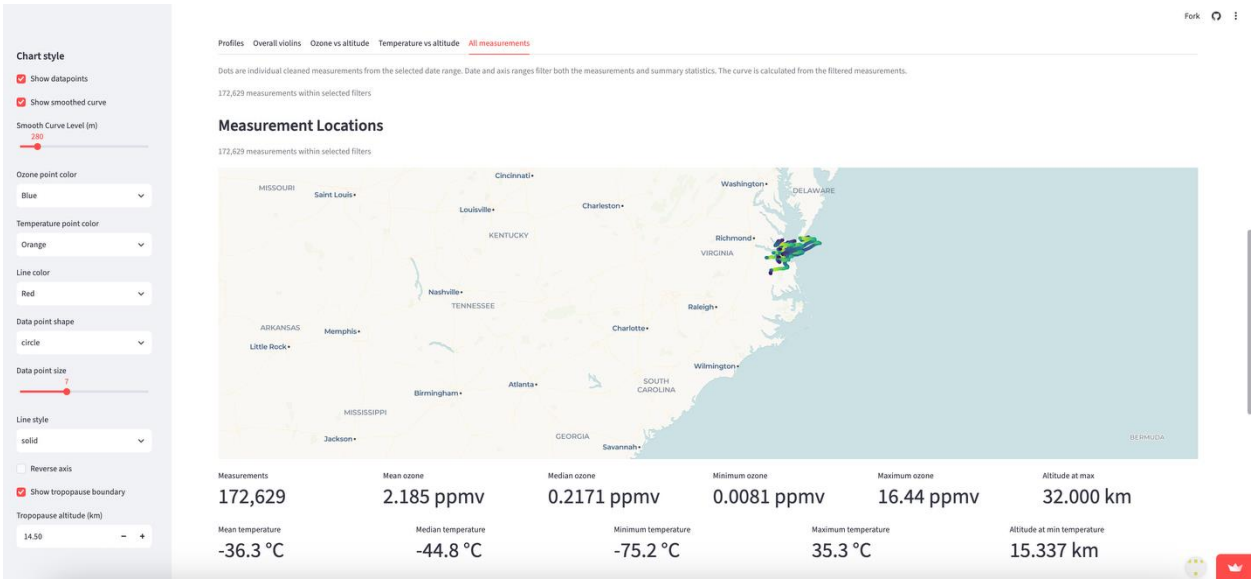


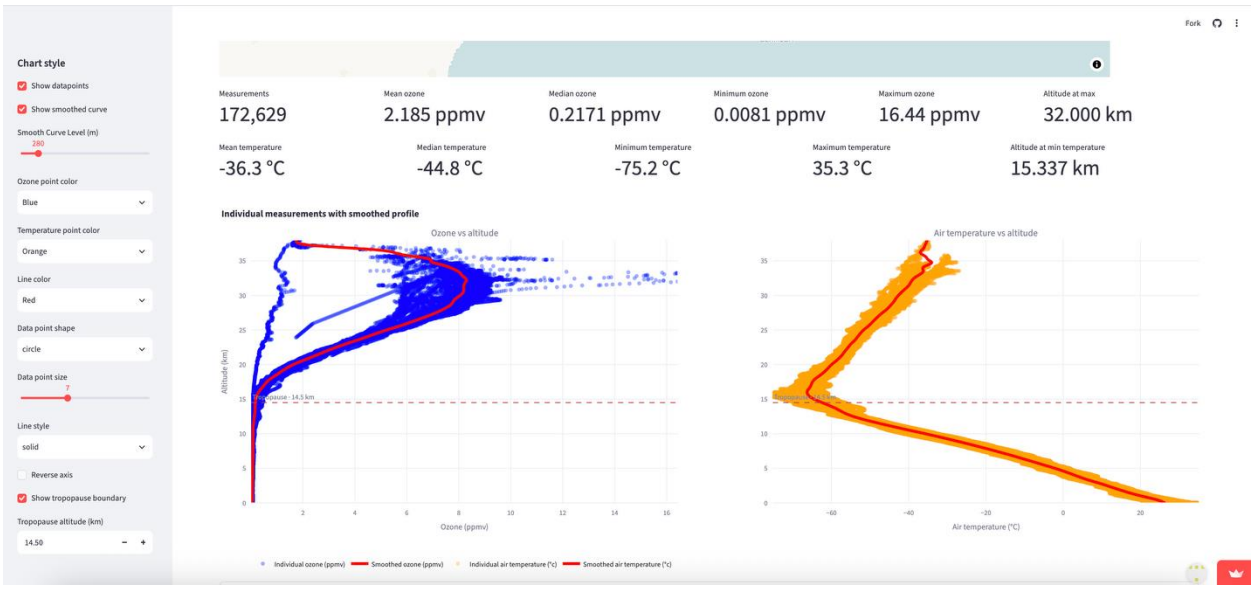
The next chart is the overall violins. This shows the distribution for altitude in km, ozone, and air temperature in Celsius. If the user hovers over the charts they can see the mean, median, max, minimum, first quartile, third quartile of the distribution. They can also hover over any part of the chart and view the measurement of the variable for that exact point.





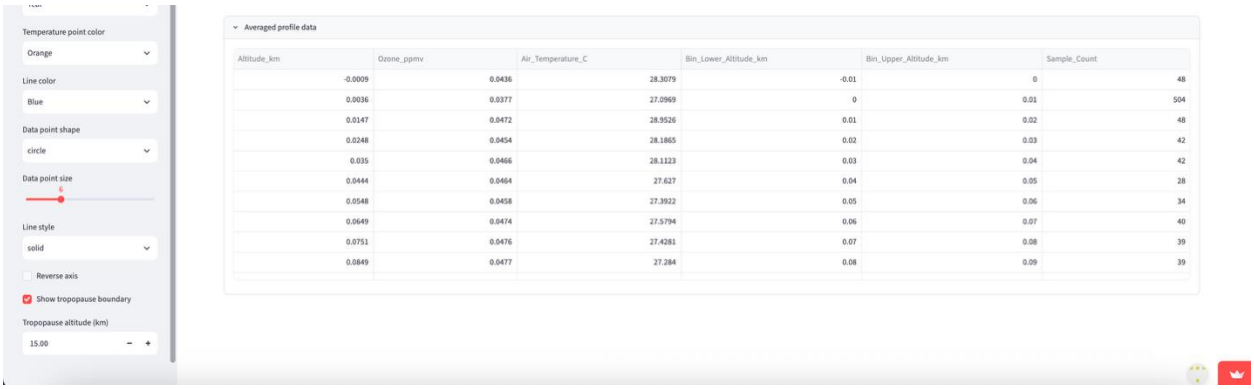
Next the user can view a chart of ozone by altitude color coded by air temperature measurements and a chart of air temperature by altitude, color coded by ozone measurements (if air temperature data is available for that dataset). Some datasets will only show a chart of ozone by altitude.





Lastly, the user can view all measurements in one chart including mean, median, max, and minimum values for all variables. It also shows a map that indicates where the measurements were recorded (This information is not available for some datasets). The bottom shows the smoothed profile charts again.

All charts have the option, in the top right corner, to view them in full screen, reset axes, pan, zoom, or download the plot as a PNG.



At the very bottom of the dashboard, there is another drop down that shows a chart with the averaged profile data.